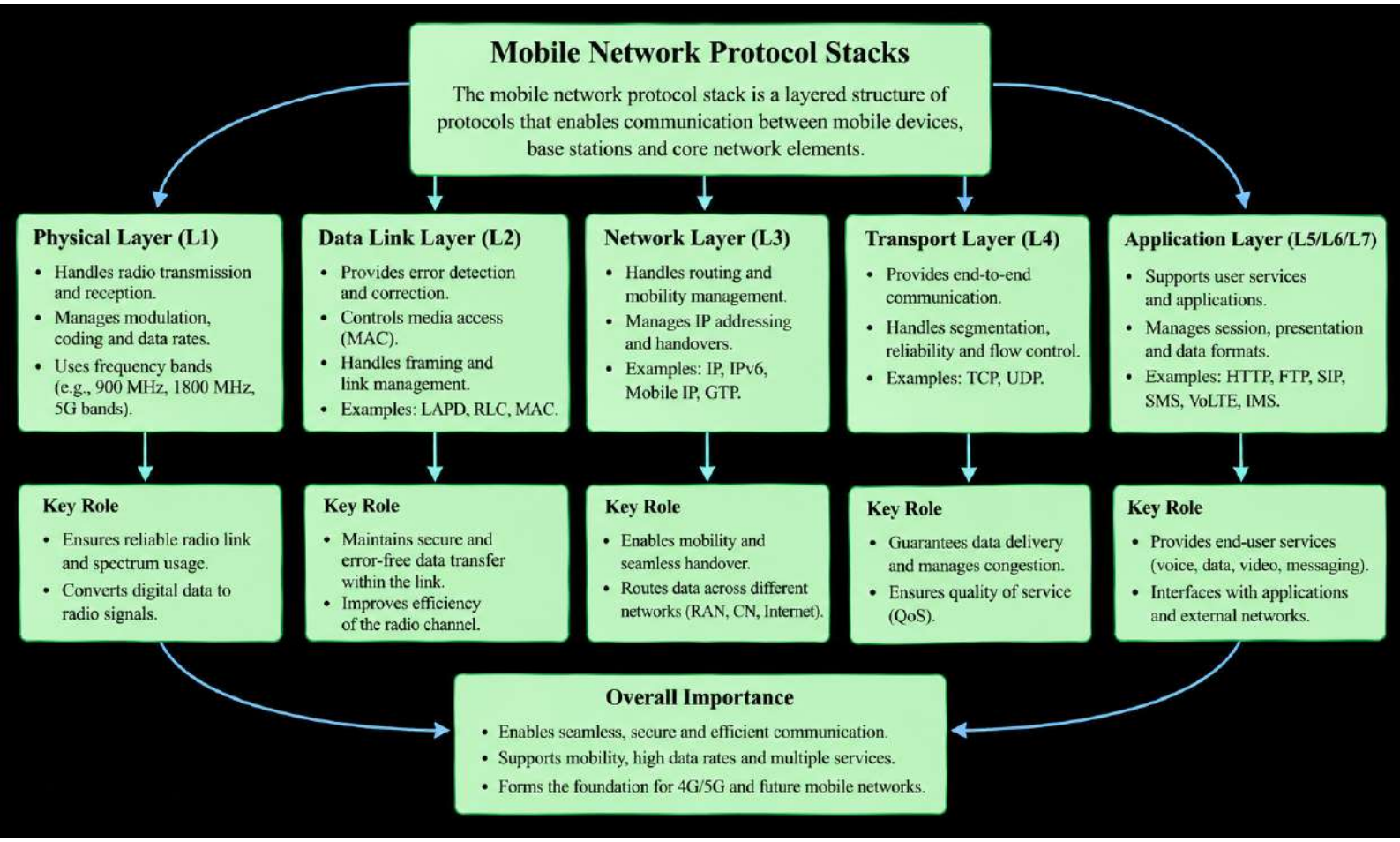

Mobile Network Protocol Stacks
The mobile network protocol stack is a layered structure of protocols that enables communication between mobile devices, base stations and core network elements.
Physical Layer (L1)
Handles radio transmission and reception.
Manages modulation, coding and data rates.
Uses frequency bands (e.g., 900 MHz, 1800 MHz, 5G bands).
Data Link Layer (L2)
Provides error detection and correction.
Controls media access (MAC).
Handles framing and link management.
Examples: LAPD, RLC, MAC.
Network Layer (L3)
Handles routing and mobility management.
Manages IP addressing and handovers.
Examples: IP, IPv6, Mobile IP, GTP.
Transport Layer (L4)
Provides end-to-end communication.
Handles segmentation, reliability and flow control.
Examples: TCP, UDP.
Application Layer (L5/L6/L7)
Supports user services and applications.
Manages session, presentation and data formats.
Examples: HTTP, FTP, SIP, SMS, VoLTE, IMS.
Key Role
Ensures reliable radio link and spectrum usage.
Converts digital data to radio signals.
Key Role
Maintains secure and error-free data transfer within the link.
Improves efficiency of the radio channel.
Key Role
Enables mobility and seamless handover.
Routes data across different networks (RAN, CN, Internet).
Key Role
Guarantees data delivery and manages congestion.
Ensures quality of service (QoS).
Key Role
Provides end-user services (voice, data, video, messaging).
Interfaces with applications and external networks.
Overall Importance
Enables seamless, secure and efficient communication.
Supports mobility, high data rates and multiple services.
Forms the foundation for 4G/5G and future mobile networks.